5. Adsorption and Ion Exchange.

Page No.
Date

Q.1 Explain ion Exchange Process?
Ans: State Principal, equilibria and rate of ion Exchange Process.

Ans: **Ion Exchange Process:**

1) The reversible exchange of ions of an electrolyte in a solution with ions of the same charge of a certain insoluble solids that thus Permit the seperation of the electrolytic solute is called Ion Exchange.
2) Ions in a solution are removed by this Process.
3) Consider the case of water softening to get the idea regarding an ion exchange Process.

Sodium Cation Exchanger:

1) Here the exchange cation is sodium.
2) When the hard water is passed through a bed of Sodium Cationite, Positively charged ions of water such as calcium and magnesium will be exchanged for the Na^+ ions of the cation exchanger.
3) The exchange reaction is reversible and Proceed as follows:

$$Ca^{2+} + 2Na^+R^- \rightleftharpoons Ca^{2+}R_2^- + 2Na^+$$

$$\underset{\text{(Solution)}}{Mg^{2+}} + \underset{\text{(Solid)}}{2Na^+R^{-}} \rightleftharpoons \underset{\text{(Solid)}}{Mg^{2+}R_2^-} + \underset{\text{(Solution)}}{2Na^+}$$

where the symbol Na^+R^- represents the sodium cationite and R^- is a complex radical of the insoluble cationite that act as a multivalent anion.

4) So the hardness of water or hard water is softened by removing hardness causing ions Ca^{2+} and Mg^{2+}
5) When the cation exchange reaches a specific capability it is regenerated with solution of NaCl.
6) The regeneration of cation Exchanger with a Sol^n of NaCl Proceeds as follows)

$$\underset{\text{(Solid)}}{CaR_2} + \underset{\text{(Sol}^n\text{)}}{2NaCl} \rightleftharpoons \underset{\text{(Solid)}}{2NaR} + \underset{\text{(Sol}^n\text{.)}}{CaCl_2}$$

Hydrogen Cation Exchanger:

1) Here the exchange cation is Hydrogen. In hydrogen cation ion exchange water softening the following reacn takes place.

$$\underset{(soln)}{Na^+} + \underset{(solid)}{H^+R^-} \rightleftharpoons \underset{(solid)}{Na^+R^-} + \underset{(soln)}{H^+}$$

$$Ca^{2+} + 2H^+R^- \rightleftharpoons Ca^{2+}R^- + 2H^+$$

$$Mg^{2+} + 2H^+R^- \rightleftharpoons Mg^{2+}R^- + 2H^+$$

$$HCO_3^- + H^+ \rightleftharpoons CO_2 + H_2O$$

2) It is evident from the above reaction that the hydrogen cation exchange process removes completely the alkanity of water treated, with the result that total dissolved solids content of water diminish.

3) Here the salt of Ca, Mg, Na etc present in water pass into a free acids since all cations are replaced by hydrogen cations.

4) The total acidity of water softned by this exchanger will be equal to the sum of anions of mineral acids Cl^-, SO_4^{2-}, NO_3^- present in the intial water, so water treated by these exchanger is not suitable for use as a feed water.

5) The hydroge cation exchanger is regenerated with 1 to 1.5% of solution of H_2SO_4

$$\underset{(solid)}{CaR_2} + \underset{(soln)}{H_2SO_4} \rightleftharpoons \underset{(solid)}{2RH} + \underset{(soln)}{CaSO_4}$$

Anion Exchange:

1) anion exchanger occurs when the fixed functional groups of the exchanger are positive and the solid is referred to as anion or anionic exchanger.

2) There is an exchange betn the electrolytes of water and insoluble solid material.
3) The anion OH^- usually used as an exchanger ion in the anion exchanger.
4) The water is passed to the bed of anion exchanger the following exchange reaction occur.

$$R^+ OH^- + Cl^- \rightleftharpoons R^+ Cl^- + OH^-$$

$$2R^+ OH^- + SO_4^{2-} \rightleftharpoons R^+ SO_4^{2-} + 2OH^-$$

5) Anion exchange are usually regenerated with a 4% soln of sodium hydroxide.

$$\underset{(solid)}{RCl} + \underset{(soln)}{NaOH} \rightleftharpoons \underset{(solid)}{ROH} + \underset{(soln)}{NaCl}$$

$$\therefore R_2SO_4 + 2NaOH \rightleftharpoons 2ROH + Na_2SO_4$$

⇒ Principal of ion Exchange:

1) An insoluble solid material used to remove ions in a solution contains fixed groups that ~~can~~ ~~carrse~~ carry c ionic charge eithe +ve or -ve in conjuction with free ions of opposite charge that can be displaced.
2) The chareges of fixed ionic groups are balanced by a diffusable ions of ~~fix~~ opposite charge, so the exchang solid as a whole electrically neutral.
3) The insoluble solid material that is used to exchange like ions with the solution is called ion exchanger.
4) Zeolites are naturally occuring clays that are capable o exchanging cations.
5) Most ion exchangers currently in large scale use are synthetic resinous materials.

Page No.
Date

→ **Ion Exchange Equilibria:**

1) The phase equilibrium relationship in ion exchange process at a given temperature is given by the expression similar in type to that of adsorption isotherm

2) The mass action relationship applied to the exchange reaction can also be used to describe the ion exchange equilibrium.

3) For example for the cation exchange reaction,

$$\underset{(soln)}{Na^+} + \underset{(solid)}{R-H^+} \rightleftharpoons \underset{(solid)}{R-Na^+} + \underset{(soln)}{H^+}$$

4) according to the ~~A~~ law of mass-action at equilibrium

$$\alpha = \frac{(\text{conc}^n\ R-Na^+)_{solid}\ (\text{Conc}^n.\ H^+)_{soln}}{(\text{conc}^n.R-H^+)_{solid}\ (\text{conc}^n.\ Na^+)_{soln}}$$

$$\therefore \alpha = \left(\frac{\text{Conc}^n.\ Na^+}{\text{Conc}^n.\ H^+}\right)_{solid} \left(\frac{\text{conc}^n\ H^+}{\text{conc}^n\ Na^+}\right)_{soln.} \quad \text{———} (1)$$

where α is the equilibrium constant for exchange rean,

5) The quantity α is therefore seen to an expression for the relative adsorptivity, in the present case it is the relative adsorptivity of Na^+ with respect to H^+

6) For most ~~engg~~ engineering application, α can be expressed as,

$$\alpha = \left(\frac{x}{x_0 - x}\right)\left(\frac{C_0 - C^*}{C^*}\right) = \left(\frac{x/x_0}{1 - x/x_0}\right)\left(\frac{1 - C^*/C_0}{C^*/C_0}\right)$$

where,

C^* is the equilibrium conc^n of Na^+ after exchang in soln

C_0 is the initial conc^n of $Na^+ + H^+$ in soln

$$\therefore \left(\frac{\text{conc}^n\ H^+}{\text{conc}^n\ Na^+}\right)_{soln} = \frac{C_0 - C^*}{C^*}, \text{ at equilibrium}$$

x is equilibrium concn of Na^+ in solid
x_0 is concn if all H^+ were replaced by Na^+

$$\left(\frac{\text{conc}^n\ Na^+}{\text{conc}^n\ H^+}\right)_{solid} = \frac{x}{x_0 - x}$$

Substituting the value in eqn (1) gives

$$\alpha = \left(\frac{x}{x_0 - x}\right)\left(\frac{C_0 - C^*}{C^*}\right) = \left(\frac{x/x_0}{1 - x/x_0}\right)\left(\frac{1 - C^*/C_0}{C^*/C_0}\right)$$

7) all the quantities in above equations are expressed as equivalents per unit volume or mass.

⇒ Rate of Ion Exchange:
Ion Exchange is a complex phenomena that involves the following steps in series and the rate of ion exchange depends on the rate of these steps.
1) Diffusion of the ions from the bulk liquid phase to the external ions of resins.
2) diffusion of the ions from the surface of the rasin ~~to~~ ~~t~~ through the resin to the site of exchange.
3) Exchange of ions at the active sites.
4) Diffusion of the released ions from inside to the surface of the resin.
5) diffusion of the released ions from the surface to the bulk of solid.

Page No.
Date

Q. differentiate betn Physical and chemical adsorption.

Physical adsorption	chemisorption.
1) Forces involved are weak	1) Forces involved are strong.
2) It is reversible Phenomena	2) It is irreversible Phenomena.
3) Heat of adsorption is small ~~or~~ of the order of heat of condensation (4 to 40 kJ/mol)	3) Heat of adsorption is large of the order of heat of reaⁿ (40 to 400 kJ/mol)
4) Non-specific in nature and entire surface is available for adsorption	4) Highly specific in nature and restricted to the active site on the surface.
5) surface coverage is complete and it may extend to multilayers	5) surface coverage is incomplete and it is limited to the unimolecular layer.
6) Quantity of substance adsorbed per unit mass of adsorbent is high	6) Quantity of substance adsorbed per unit mass of adsorbent is low.
7) Activation energy is low nearly negligible	7) activation energy is high corresponding to reaⁿ
8) Adsorption of O_2, H_2 gases ~~are~~ on charcoal	8) Adsorption of H_2 gas on nickel metal.

Page No.
Date

Q. Explain the characteristics of adsorbents and mention diff. types of adsorbents with their specific applications.

Ans: **Characteristics of Adsorbent:**

1) Adsorbent solids are usually used in granular form with size ranging from 12 mm in diameter to 50 μm
2) They must possess the large surface area per unit mass.
3) They should have preferential ability to take up adsorbate.
4) These are usually highly porous material.
5) They should be free flowing for ease of handling.
6) They should offer a low pressure drop for flow of fluid when used in a fixed bed.
7) They should have adequate strength and hardness.

Adsorbents in general use:

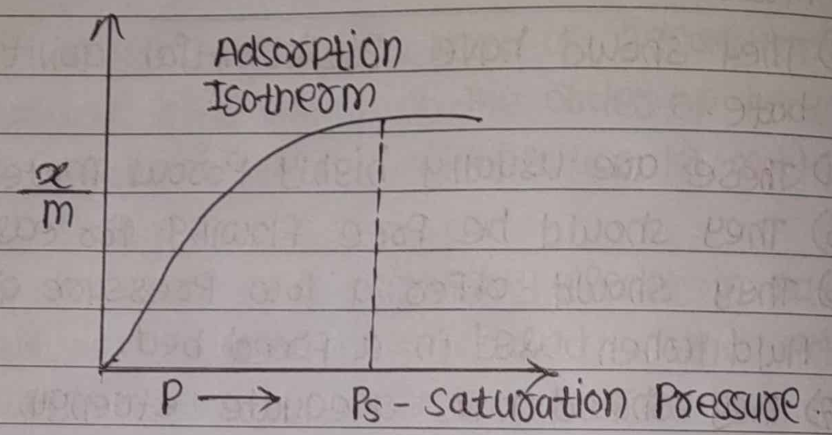
1) Activated carbon - it is used for decolourising of sugar, drugs, chemicals, water purification etc.
2) Activated alumina - it is used for recovery of solvent vapors from gas mixtures, gas masks etc.
3) Silical gel is used for dehydration of air and other gases
4) Molecular seives are used for dehydration of gases and liquids, seperation of gas and liquid hydrocarbon mixtures.
5) Bone char is used for ~~er~~ refining of sugar
6) Activated clays are used for decolourising petroleum product
7) ~~Fullers~~ Fuller's earth are useful for decolourising lubricating oils, kerosene, gasoline and vegetable oils.
8) Charcoal is used for removal of gases like SO_2, NH_3, etc
9) Nickel is used in hydrogenation of vegetable oils - H_2 adsorption.

Page No.
Date

Q. Explain different adsorption isotherm.

Ans: Adsorption isotherm:

1) An adsorption isotherm is a graph that represents the variation in the amount of adsorbate (x) adsorbed on the surface of the adsorbent with the damage in pressure at a constant temperature.



(1) Freundlich Adsorption Isotherm:

1) Freundlich adsorption gives the variation in the quantity of gas adsorbed by a unit mass of solid adsorbent with the change in pressure of the system for a given temperature.

2) The expression for the freundlich isotherm can be represented by the following equation:

$$\frac{x}{m} = KP^{1/n} \quad , \quad n > 1 \quad \left(\because \frac{x}{m} = V\right)$$

where x is mass of the gas adsorbed
m is the mass of adsorbent
P is the pressure.
and n is a constant which depends upon the nature of adsorbent at a given temperature

3) Taking logarithm on both sides of the eqn,

$$\log \frac{x}{m} = \log K + \frac{1}{n} \log P$$

4) A Plot of $\log \frac{x}{m}$ versus $\log P$ should give a straight line.

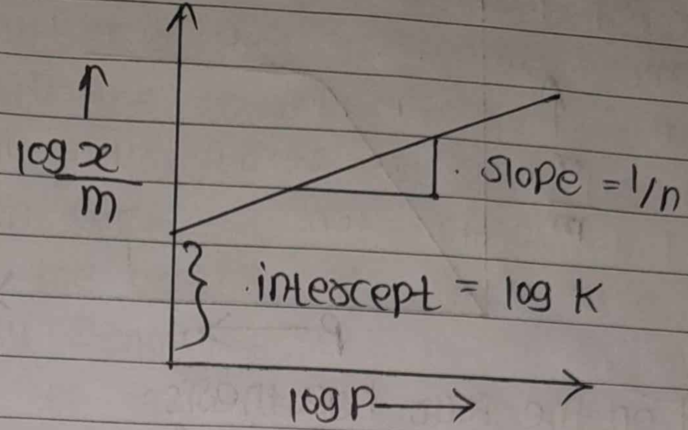


fig Freundlich Adsorption isotherm

5) Freundlich equation of an adsorption from soln,

$$T = \frac{x}{m} = kC^{1/n}$$

where, P and C is the equilibrium Pressure and Concn

(2) langmuir Adsorption isotherm:

1) The langmuir adsorption isotheorm Predict linear adsorPtion ~~densities~~ at low adsorption densities and a maximum surface coverage at higher solute metal Concn.

2) The langmuir adsorption has the form:

$$\theta = \frac{KP}{1+KP}$$

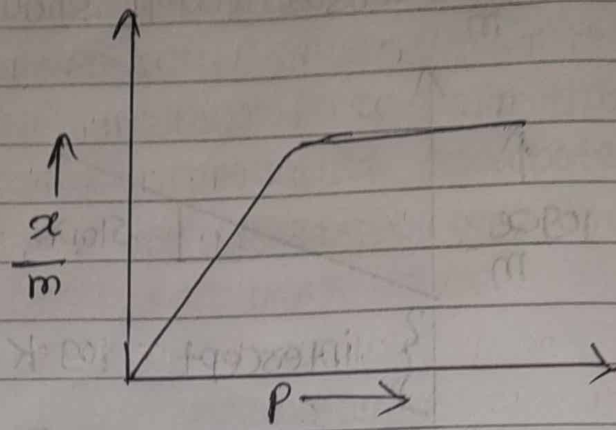
where, θ = fraction of the surface covered by adsorbed molecule.

K is equilibrium constant known as the adsorption coefficient

$$\therefore K = \frac{K_a}{K_d} = \frac{\text{rate constant for adsorption}}{\text{rate constant for desorption}}.$$

P is the Pressure.

3) The langmuir adsorption is applicable for monolayer adsorption onto a homogeneous surface when no interaction occurs betn adsorbed species.



It is based on the four hypothesis:

1) The surface of the adsorption is uniform that is all the adsorption sites are equal.
2) Adsorbed molecules do not interact
3) All adsorption occur through same mechanism.
4) At the maximum adsorption only a monolayer is formed.

Q. Explain Breakthrough Curve.

Ans: 1) When binary solution containing strongly adsorbed solute at concentration Y_0 mass solute / mass solvent flows through a relatively deep bed of adsorbent initially free from adsorbate.

2) The topmost layer of adsorbent which first comes in contact with the strong solution fed, at first adsorbs solute rapidly and effectively and whenever a little amount of it left the solution unadsorbed then removed by the layers of adsorbent in the lower part of the bed.

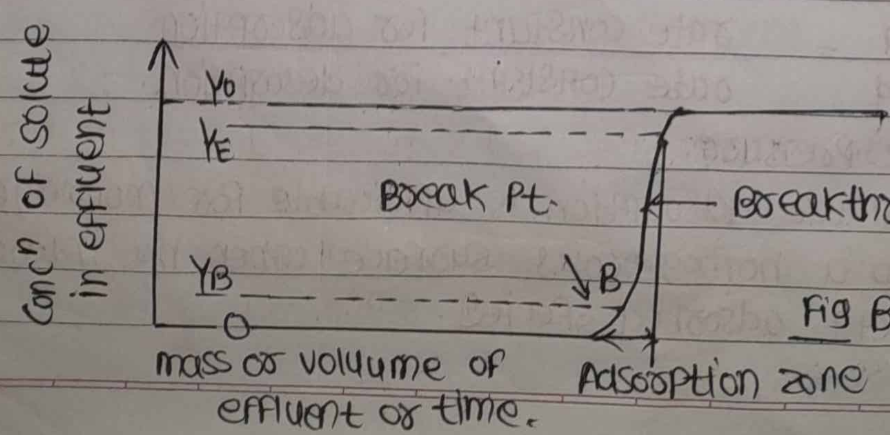


Fig Breakthrough Curve.

Page No.
Date

3) The Part of the effluent concentration curve betn Pt. B and E is termed as the Breakthrough curve.
4) The Portion of a Plot of effluent concn as a Function of time betn the breakpoint and time corresponding to Practical saturation of the bed is termed as the breakthrough curve.
5) Generally the breakthrough curve is shaped and it is often nearly symmetrical.
6) The slope of the breakthrough curve is a function of the rate of adsorption.
7) The higher rate of adsorption, the steeper the break through curve
8) If the adsorption Process is instantaneous, the adsorption zone would be of an infinitesimal width, and the breakthrough curve would be straight vertical line from 0 to Y_0 when all the adsorbent solid was saturated.
9) The breakpoint - The point in time at which solute is first noticed in the effluent is often taken as a solute concn of $Y_B = 0.05 Y_0$
10) Similarly the point at which the bed is Practically saturated is often taken as a solute concn of $Y_E = 0.95 Y_0$.
10) For the symmetrical curve the ideal adsorption time is time when $Y = 0.5 Y_0$

Q. Explain the minimum adsorbent requirement for infinite stage in adsorption isotherm.

Ans: Determination of minimum adsorbent requirement:

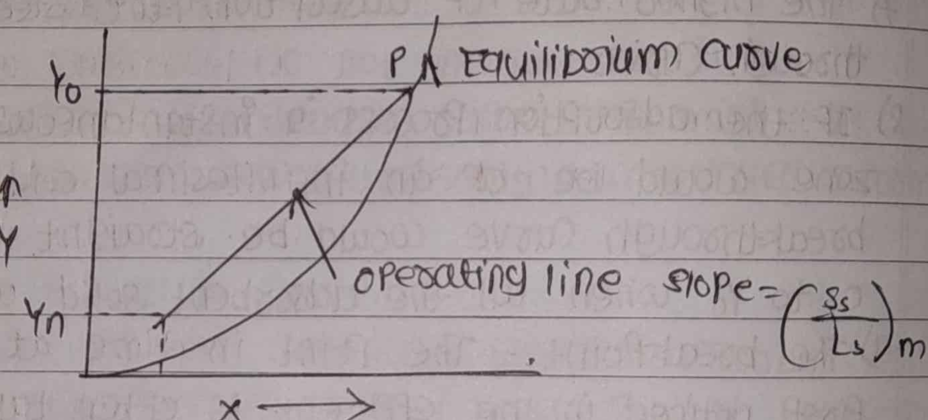
1) The minimum adsorbent ratio requiring an infinite number of stages for ~~a~~ the desired changes in concn of the solute in the solution corresponds to the operating line of the largest slope that touches the equilibrium curve.

① For the equilibrium Curve - straight or Concave upward

1) on X-Y Plot locate Pt Q (X_{n+1}, Y_n) and through Y_0 draw a line Parallel to X-axis which will cut the equilibrium curve at P.

2) Join PQ, which is the operating line for the minimum adsorbent to solvent ratio.

3) The ~~soli~~ slope of operating line gives the value of $(S_s/L_s)_m$, so measure the slope and then calculate the minimum ~~adsorbeent~~ amount of adsorbent required (as L_s is known)



② For the equilibrium curve- concave downward:

1) In this case, locate the Point Q (X_{n+1}, Y_n).

2) Through the Point Q, draw a line which will be a tangent to the curve and measure the slope of this line to get $(S_s/L_s)_m$ and then calculate the minimum adsorbent required from the value of slope and known value of L_s.